TASK 2 — STEP-BY-STEP GUIDE

IDENTIFY USE CASES · WORKED EXAMPLES & STUDENT PRACTICE

Course: Geo Programming | Week 5 | Suggested time: 15–25 minutes | Individual work

1. Learning Outcomes

Identify actors and goals; derive at least six use cases; describe actor, precondition, main flow and expected result; model error flows; and produce a use-case table or diagram linked to Task 1 requirements.

2. Problem Statement — Review Before Starting

For academic year 2026–2027, a school uses an iOS application to manage students, teachers, subjects and enrollments. Staff can browse records, add/edit students, assign subjects to teachers, enroll students, enter/update marks (0–10), and inspect student results. Duplicate enrollment in the same subject and academic year is not allowed. A teacher may teach several subjects, and a subject may have several teachers.

Important: Task 1 requirements are your input. Focus on WHAT users achieve, not Swift class names or screen implementation.

3. Guided Procedure

Step 1 — Identify Actors and Goals (2 minutes)

An actor is a role interacting with the system. The specification explicitly says “staff”; specialized roles below are proposed assumptions, not stated facts.

Example	Actor	Goal	Status
1	School staff	Browse student records	Specified
2	School staff	Enroll a student in a subject	Specified
3	Academic coordinator	Assign a subject to a teacher	Assumed specialization of staff

YOUR TURN: Propose 3–5 additional actor–goal pairs. Mark every new role as specified or assumed.

No.	Actor / role	Goal	Specified / assumed?
1	Academic Administrator (specialization of staff)	Add and edit student records	Assumed
2	Teacher / Grader (specialization of staff)	Enter and update student marks (0–10)	Assumed
3	School staff	Inspect a student’s subjects, academic years and marks	Specified
4	School staff	Browse subjects, teachers and enrollments	Specified (“browse records”)
5	Academic Administrator	Enroll a student while preventing duplicate enrollments	Role assumed; goal specified

Step 2 — Convert Functional Requirements into Use Cases (3 minutes)

Start with an FR from Task 1, then express the user goal as a verb + object. One use case may cover multiple related requirements.

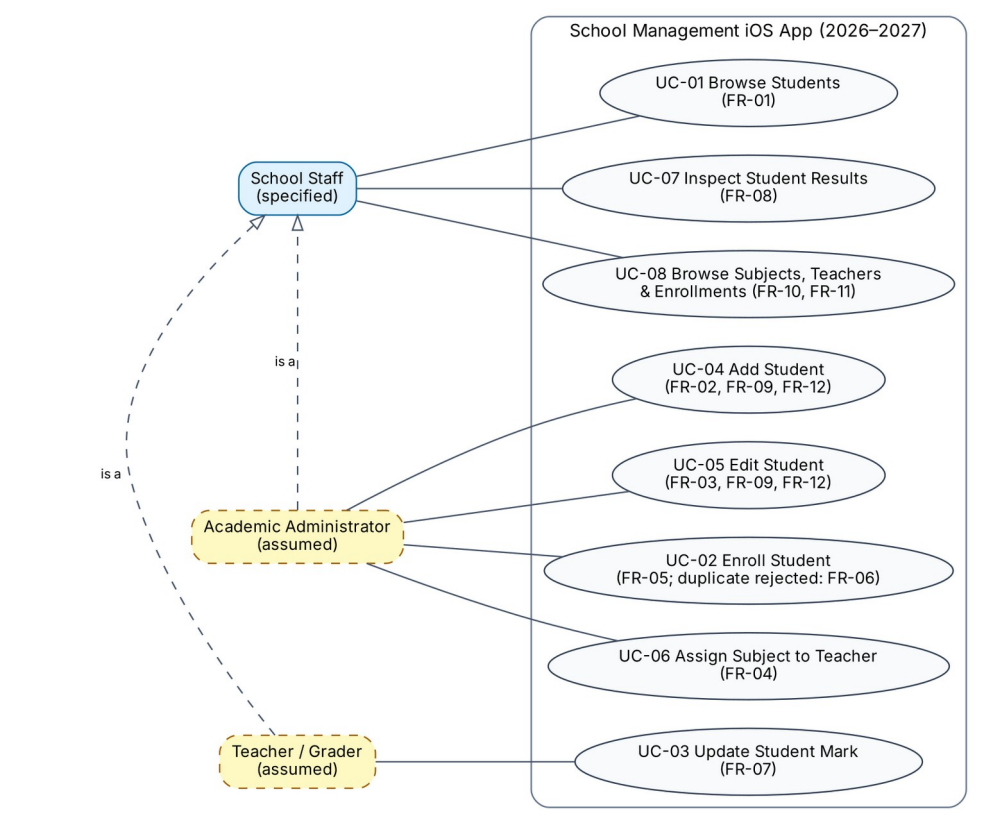
Example	Task 1 FR (illustrative)	Use case
1	FR-01: Display student records	UC-01 — Browse Students
2	FR-04: Enroll a student	UC-02 — Enroll Student
3	FR-06: Update a mark	UC-03 — Update Student Mark

YOUR TURN: Derive 3–5 additional use cases from your own Task 1 FRs. Use unique UC identifiers.

FR ID(s)	Use-case ID	Verb + object name
FR-02, FR-09, FR-12	UC-04	Add Student
FR-03, FR-09, FR-12	UC-05	Edit Student
FR-04	UC-06	Assign Subject to Teacher

FR-08	UC-07	Inspect Student Results
FR-10, FR-11	UC-08	Browse Subjects, Teachers and Enrollments

Use-Case Diagram (UC-01 – UC-08, traced to Task 1 FRs)



Solid actor = specified (School Staff). Dashed yellow actors = assumed specializations of School Staff. FR numbers refer to the real Task 1 table (FR-10 – FR-12 are proposed additions).